

ECE 315 – Signals and Systems, Spring 2026

Midterm Exam I

March 6, 2026

Exam duration: 50 minutes

INSTRUCTIONS

1. This is a **closed-book** examination.
2. The exam has **four** questions and comprises **seven** pages.
3. Show all work for full credit.
4. Students may use scientific calculators, but wireless communication is prohibited. Other electronic devices (e.g., computers, phones, tablets, digital notebooks, etc.) are not allowed.
5. The mathematical formulas below may be helpful.

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$$

$$\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$$

$$\sin(x + y) = \sin(x) \cos(y) + \cos(x) \sin(y)$$

$$\cos(x + y) = \cos(x) \cos(y) - \sin(x) \sin(y)$$

$$\sin(x) \cos(y) = \frac{1}{2}(\sin(x - y) + \sin(x + y))$$

$$\sum_{n=0}^{N-1} \alpha^n = \frac{1-\alpha^N}{1-\alpha} \quad \text{for } \alpha \neq 1$$

$$\sum_{n=0}^{\infty} \alpha^n = \frac{1}{1-\alpha} \quad \text{for } |\alpha| < 1$$

Question 1

(4pts) Consider the following complex discrete-time signal:

$$x[n] = \sin\left(\frac{\pi}{4}n\right) + \cos\left(\frac{9\pi}{4}n\right).$$

- (i) Compute the energy and average power of $x[n]$;
- (ii) Determine whether $x[n]$ is periodic and if so, compute its fundamental period.

Question 2

(4pts) Consider the following discrete-time systems with input $x[n]$ and output $y[n]$:

- $y[n] = (x[n - 1])^2$;
- $y[n] = \min(-1, x[1 - n])$.

Determine whether they are memoryless, invertible, causal, time-invariant, and linear. Justify your answer.

Question 3

(6pts) Consider the following continuous-time linear time-invariant system that is represented by an impulse response $h(t)$:

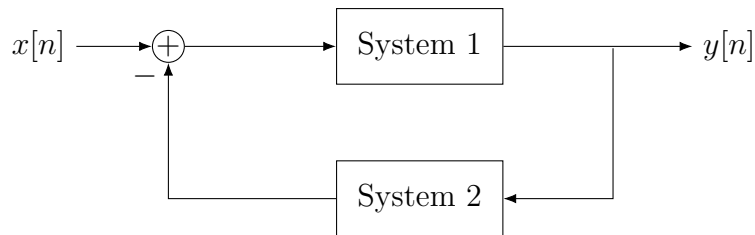
$$h(t) = \begin{cases} 2^{-t} & \text{when } 0 \leq t \leq 1 \\ 1 & \text{when } -1 \leq t \leq 0 \\ 0 & \text{otherwise.} \end{cases}$$

- (i) Determine whether it is memoryless, causal, and stable. Justify your answer;
- (ii) Compute the system's output for the input $x(t) = u(t - 1)$.

Question 4

(6pts) Consider the following feedback system (w/ input $x[n]$ and output $y[n]$) that consists of two subsystems:

- System 1 with impulse response $h_1[n] = \delta[n]$;
- System 2 with impulse response $h_2[n] = \frac{1}{3}\delta[n - 1]$.



- Find a difference equation that describes the input-output relationship;
- Assume the condition of initial rest holds. Compute the system's impulse response. Check whether it is invertible and if so, compute the impulse response of the inverse system.

